

GRE Review

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1 Problems

Let x, y be functions of t . Then $\frac{dy}{dt} = \frac{dx}{dt} \cdot \frac{dy}{dx} \rightarrow \frac{dy}{dx} = \frac{dy}{dt} / \frac{dx}{dt}$. We compute

$$\begin{aligned} \frac{d}{dt} \frac{dy}{dx} &= \frac{d}{dt} \left(\frac{dy}{dt} / \frac{dx}{dt} \right) = \frac{\frac{d^2y}{dt^2} \cdot \frac{dx}{dt} - \frac{dy}{dt} \cdot \frac{d^2x}{dt^2}}{\left(\frac{dx}{dt} \right)^2} \\ &= \frac{\frac{d}{dt} \frac{dy}{dx} = \frac{dx}{dt} \cdot \frac{d^2y}{dx^2}}{\frac{d^2y}{dt^2} \cdot \frac{dx}{dt} - \frac{dy}{dt} \cdot \frac{d^2x}{dt^2}} \\ &\rightarrow \frac{d^2y}{dx^2} = \frac{\frac{d^2y}{dt^2} \cdot \frac{dx}{dt} - \frac{dy}{dt} \cdot \frac{d^2x}{dt^2}}{\left(\frac{dx}{dt} \right)^3} \end{aligned}$$

Consider when $x = t^2 + 2t$ and $y = 3t^4 + 4t^3$. Then $x' = 2t + 2, x'' = 2, y' = 12t^3 + 12t^2, y'' = 36t^2 + 24t$. At $t = 2$, we have $x' = 6, x'' = 2, y' = 144, y'' = 192$, so $\frac{d^2y}{dx^2} = \frac{192 \cdot 6 - 144 \cdot 2}{6^3} = 4$.

Suppose $x = r \cos \theta$ and $y = r \sin \theta$. Then $\frac{dx}{d\theta} = \frac{dr}{d\theta} \cdot \cos \theta + r \cdot (-\sin \theta)$ and $\frac{dy}{d\theta} = \frac{dr}{d\theta} \cdot \sin \theta + r \cdot \cos \theta$, so

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \cdot \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cdot \cos \theta - r \sin \theta}$$

Further,

$$\begin{aligned} f(r, \theta) &= \frac{d}{d\theta} \left(\frac{dr}{d\theta} \cdot \sin \theta + r \cos \theta \right) = \frac{d^2r}{d\theta^2} \cdot \sin \theta + \frac{dr}{d\theta} \cdot \cos \theta + \frac{dr}{d\theta} \cdot \cos \theta - r \sin \theta \\ g(r, \theta) &= \frac{d}{d\theta} \left(\frac{dr}{d\theta} \cdot \cos \theta - r \sin \theta \right) = \frac{d^2r}{d\theta^2} \cdot \cos \theta - \frac{dr}{d\theta} \cdot \sin \theta - \frac{dr}{d\theta} \cdot \sin \theta - r \cos \theta \\ &\rightarrow \frac{d}{d\theta} \frac{dy}{dx} = \frac{f(r, \theta) \cdot \left(\frac{dr}{d\theta} \cdot \cos \theta - r \sin \theta \right) - g(r, \theta) \cdot \left(\frac{dr}{d\theta} \cdot \sin \theta + r \cos \theta \right)}{\left(\frac{dr}{d\theta} \cdot \cos \theta - r \sin \theta \right)^2} \\ &\rightarrow \frac{d^2y}{dx^2} = \left(\frac{d}{d\theta} \frac{dy}{dx} \right) / \frac{dx}{d\theta}. \end{aligned}$$

Concavity of parametric curve:

$$\frac{d^2y}{dx^2} = \frac{d^2y}{dt dx} / \frac{dx}{dt}.$$

Again: $\frac{dy}{dx} = \frac{12t^3+12t^2}{2t+2} = 6t^2$, so $\frac{d^2y}{dx^2} = 12t/(2t+2) = 6t/(t+1)$, so at $t = 2$, we have $\frac{d^2y}{dx^2} = 4$.

We have

$$\begin{aligned} y' + xy = x &\rightarrow e^{x^2/2}y' + xe^{x^2/2}y = xe^{x^2/2} \rightarrow (e^{x^2/2}y)' = xe^{x^2/2} \\ &\rightarrow e^{x^2/2}y = e^{x^2/2} + C \rightarrow y = 1 + Ce^{-x^2/2}. \end{aligned}$$

Then regardless of C , we have $\lim_{x \rightarrow -\infty} y = 1$.

We have

$$\begin{aligned} y' + xy = x &\rightarrow y' = x(1-y) \rightarrow y'/(y-1) = -x \\ \rightarrow \log|y-1| &= -x^2/2 + C \rightarrow y-1 = Ce^{-x^2/2} \rightarrow y = 1 + Ce^{-x^2/2}. \end{aligned}$$

How many positive solutions to $\cos 97x = x$? There are $1/(2\pi/97) = 97/2\pi \approx$ periods.

Find all points where f is continuous and where f is differentiable, for

$$f(x) = \begin{cases} 3x^2 & x \in \mathbb{Q}, \\ -5x^2 & x \notin \mathbb{Q}. \end{cases}$$

Suppose f is continuous at $a \in \mathbb{R}$. Let (x_n) be a sequence of rationals with $x_n \rightarrow a$. Then $f(x_n) \rightarrow f(a)$, so $3x_n^2 \rightarrow 3a^2 = f(a)$. Similarly, for (x_n) irrational, we have $f(x_n) \rightarrow f(a) = -5a^2$, so $a = 0$. $\lim_{x \rightarrow 0} f(x) = 0 = f(0)$ by Squeeze lemma.

Differentiability implies continuity, so f can only be differentiable at $x = 0$. Indeed, $\lim_{h \rightarrow 0} \frac{3h^2}{h} = \lim_{h \rightarrow 0} \frac{-5h^2}{h} = 0$.

Let $g(x, y, z) = 3x^2y + z$. Find the directional derivative of g in $\langle 1, 2, 3 \rangle$ at $p = (0, 0, \pi)$.

Ans: Compute $\nabla g = \langle 6xy, 3x^2, 1 \rangle$, so $\nabla g(p) = \langle 0, 0, 1 \rangle$. Then the directional is $3/\sqrt{14} \approx 0.8$.

Let $f \in C^0, g \in C^3$ such that $g(x) = \int_0^x f(y)(y-x)dy$. Then for what maximal n is it guaranteed that $f \in C^n$?

Ans: Have $g(x) = \int_0^x f(y)ydy - x \int_0^x f(y)dy$, so $g'(x) = f(x)x - \int_0^x f(y)dy -$

$xf(x) = -\int_0^x f(y)dy$, so $g''(x) = -f(x)$, so $f \in C^1$.

Let $a, b > 0$. Compute $\int_0^\infty \frac{e^{ax} - e^{bx}}{(1+e^{ax})(1+e^{bx})} dx$.

We have

$$\begin{aligned} \int_0^\infty \frac{e^{ax} - e^{bx}}{(1+e^{ax})(1+e^{bx})} dx &= \int_0^\infty \frac{1}{1+e^{bx}} - \frac{1}{1+e^{ax}} dx \\ &= \int_0^\infty \frac{e^{ax}}{1+e^{ax}} - \frac{e^{bx}}{1+e^{bx}} dx = \left(\frac{1}{a} \log(1+e^{ax}) - \frac{1}{b} \log(1+e^{bx}) \right) \Big|_0^\infty \\ &= 0 - (1/a \log(2) - 1/b \log(2)) = (1/b - 1/a) \log(2) = \frac{a-b}{ab} \log 2. \end{aligned}$$

Compute $1/x \leq 1/\sqrt{x}$ and $\log 1 \leq 2\sqrt{1}$, so $\log x \leq 2\sqrt{x}$ for all $x \geq 1$.

(i) is true by Squeeze lemma. Pick $x_n = 1/(2n)$ and let $f : (0, 1) \rightarrow \mathbb{R}$ be given by $f(x) = 1/x$. Then $f(x_n) = (2n)$, which is not Cauchy.

Claim: Let $f : A \rightarrow \mathbb{R}$ be continuous at $a \in A$ and $x_n \rightarrow a$ be a sequence in A . Then $\lim_{n \rightarrow \infty} f(x_n) = f(a)$.

Proof: Let $\epsilon > 0$. Pick by continuity of f at a some $\delta > 0$ such that $x \in V_\delta(a) \cap A \rightarrow f(x) \in V_\epsilon(f(a))$. Pick by definition of limit some $N \in \mathbb{Z}^+$ such that $n \geq N \rightarrow x_n \in V_\delta(a)$. Then for $n \geq N$, we have $f(x_n) \in V_\epsilon(f(a))$, ie $\lim_{n \rightarrow \infty} f(x_n) = f(a)$.

Claim: Let $x_n \subseteq A$ be Cauchy and $f : A \rightarrow \mathbb{R}$ be uniformly continuous. Then $f(x_n)$ is Cauchy.

Proof: Let $\epsilon > 0$. Pick $\delta > 0$ such that for $x, y \in A$ that $|x - y| < \delta \rightarrow |f(x) - f(y)| < \epsilon$. Pick $N \in \mathbb{Z}^+$ such that $n, m \geq N \rightarrow |x_n - x_m| < \delta$. Then for $n, m \geq N$, we have $|f(x_n) - f(x_m)| < \epsilon$, ie $f(x_n)$ is Cauchy. It follows $\lim_{n \rightarrow \infty} f(x_n)$ exists.

Now $\lim_{n \rightarrow \infty} x_n = 0$ exists, so x_n is Cauchy. Then $g(x_n)$ is Cauchy, so $\lim_{n \rightarrow \infty} g(x_n)$ exists.

Compute $L(\theta_0) = \int_0^{\theta_0} \sqrt{(-r \sin \theta)^2 + (r \cos \theta)^2 + 1} d\theta = \theta_0 \sqrt{r^2 + 1}$ and $D(\theta_0) = \sqrt{(r \cos \theta_0)^2 + (r \sin \theta_0)^2 + \theta_0^2} = \sqrt{r^2 + \theta_0^2}$. Then $L(\theta_0)^2 = \theta_0^2(r^2 + 1)$ and $D(\theta_0)^2 = r^2 + \theta_0^2$, so $L(\theta_0)^2 = (D(\theta_0)^2 - r^2)(r^2 + 1)$. If $r = 5, L(\theta_0) = 26$, then $D(\theta_0) = \sqrt{51}$.

Sps $(I - A)^k = 0$ for some $k \in \mathbb{Z}^+$. Sps $v \neq 0$ such that $Av = 0$. Then $(I - A)^k v = (I - A)^{k-1} v = v$, so $v = 0$, absurd.

Let a tank with volume V have initial salt q_0 . Assume salt of concentration

μ enters, leaves at a rate r , and the tank is well-mixed. Then find q as a function of time t .

Ans: Have $\frac{dq}{dt} = \mu r - \frac{q}{V} r$ and $q(0) = q_0$. Then $\frac{dq}{\frac{q}{V} - \mu} = -r dt \rightarrow V \log |q/V - \mu| = -rt + C \rightarrow q/V - \mu = C e^{-rt/V} \rightarrow q = V\mu + C e^{-rt/V}$, so $q = V\mu + (q_0 - V\mu)e^{-rt/V}$.

Consider $V = 100L$, $\mu = 0.02g/L$, $r = 4L/m$. Then $q(4m) = (100L)(0.02g/L) + (3g - 2g)e^{-(4L/m)(100m)/100L} = 2 + e^{-4}$.

Let $A, B \neq \emptyset$ be subsets of $\mathbb{R}$. Define $A \cdot B = \{x \cdot y : x \in A, y \in B\}$. If $A, B \neq \emptyset$ are bounded and $\sup A > \sup B$, then $\sup A \cdot B = ?$

Ans: Claim $\sup A \cdot B = \max\{\sup A \cdot \sup B, \sup A \cdot \inf B, \inf A \cdot \sup B, \inf A \cdot \inf B\}$. Sps FTSOC $\sup A \cdot \inf B > \sup A \cdot \sup B, \inf A \cdot \sup B, \inf A \cdot \inf B$. Then $\sup A \cdot (\inf B - \sup B) > 0$, so $\sup A < 0$ and $(\sup A - \inf A) \cdot \inf B > 0$, so $\inf B > 0$. Then $0 > \sup A > \sup B > \inf B > 0$, absurd.

Compute the flux of $F = \langle x, y, z \rangle$ through the upper-half hemisphere of the unit sphere, oriented outwards from the origin.

Ans:

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{A} &= \int_{\varphi=0}^{\pi/2} \int_{\theta=0}^{2\pi} \langle \cos \theta \sin \varphi, \sin \theta \sin \varphi, \cos \varphi \rangle \cdot \langle \cos \theta \sin \varphi, \sin \theta \sin \varphi, \cos \varphi \rangle d\theta d\varphi \\ &= \int_{\varphi=0}^{\pi/2} \int_0^{2\pi} d\theta d\varphi = \iint_S dA = 2\pi. \end{aligned}$$

Let $D \subseteq \mathbb{R}^2$ be subset of the $x-y$ plane and let $z = f(x, y)$. Assume $S = f(D)$ is oriented upwards and let $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a vector field. Then compute the flux of F through S .

Let $R = [x_0, x_1] \times [y_0, y_1]$ and let $ax + by + cz = d$ with $c \neq 0$ be a plane. Let $z = f(x, y) = \frac{d-ax-by}{c}$ and $p_{ij} = (x_i, y_j, f(x_i, y_j))$. Note that

$$\begin{aligned} p_{00} + p_{11} &= (x_0, y_0, (d - ax_0 - by_0)/c) + (x_1, y_1, (d - ax_1 - by_1)/c) \\ &= (x_0 + x_1, y_0 + y_1, (2d - a(x_0 + x_1) - b(y_0 + y_1))/c) \\ p_{01} + p_{10} &= (x_0, y_1, (d - ax_0 - by_1)/c) + (x_1, y_0, (d - ax_1 - by_0)/c) \\ &= (x_0 + x_1, y_0 + y_1, (2d - a(x_0 + x_1) - b(y_0 + y_1))/c) \\ &\rightarrow p_{00} + p_{11} = p_{01} + p_{10}, \end{aligned}$$

so $P = (p_{00}, p_{01}, p_{11}, p_{10})$ is a parallelogram. We compute the area of P by

$$\begin{aligned}
[P] &= |(p_{01} - p_{00}) \times (p_{10} - p_{00})| \\
&= |\langle 0, y_1 - y_0, -b(y_1 - y_0)/c \rangle \times \langle x_1 - x_0, 0, -a(x_1 - x_0)/c \rangle| \\
&= (y_1 - y_0)(x_1 - x_0) \sqrt{\left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 + 1} \\
&= [R] \frac{\sqrt{a^2 + b^2 + c^2}}{c}.
\end{aligned}$$

In general, if $z = f(x, y)$, we have $dS = dA \cdot \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 + 1}$. More generally,

$$d\vec{S} = dx dy \left\langle -\frac{\partial f}{\partial x}, -\frac{\partial f}{\partial y}, 1 \right\rangle.$$

2 Iterated Integrals

Let C be simple, piecewise smooth, positively oriented, closed curve enclosing region D . Let P, Q have continuous partial derivatives on D . Then

$$\int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Justification: It is enough to show $\int_C P dx = - \iint_D \frac{\partial P}{\partial y} dy$ and $\int_C Q dy = \iint_D \frac{\partial Q}{\partial x} dA$. Consider a rectangle $R = [x_0, x_1] \times [y_0, y_1]$. We compute

$$\begin{aligned}
\iint_R \frac{\partial P}{\partial y} dA &= \int_{x_0}^{x_1} \int_{y_0}^{y_1} \frac{\partial P}{\partial y} dy dx = \int_{x_0}^{x_1} P(x, y_1) - P(x, y_0) dx \\
&= - \int_C P dx.
\end{aligned}$$

Claim: $\text{div curl } \vec{F} = 0$.

Proof: We have

$$\begin{aligned}
\text{div curl } \vec{F} &= \text{div } \det \begin{bmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{bmatrix} = \text{div} \left(i \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) - j \left(\frac{\partial F_z}{\partial x} - \frac{\partial F_x}{\partial z} \right) + k \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \right) \\
&= \left(\frac{\partial^2 F_z}{\partial x \partial y} - \frac{\partial^2 F_y}{\partial x \partial z} \right) - \left(\frac{\partial^2 F_z}{\partial y \partial x} - \frac{\partial^2 F_x}{\partial y \partial z} \right) + \left(\frac{\partial^2 F_y}{\partial z \partial x} - \frac{\partial^2 F_x}{\partial z \partial y} \right) = 0.
\end{aligned}$$

We have

$$\oint_C \vec{F} \cdot d\vec{s} = \iint_D (\text{curl } \vec{F}) \cdot \vec{k} dA.$$

Indeed,

$$\begin{aligned}\iint_D \operatorname{curl} \vec{F} \cdot \vec{k} dA &= \iint_D \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) dA = \int_C F_x dx + F_y dy \\ &= \int_C \vec{F} \cdot d\vec{s}.\end{aligned}$$

$$\begin{aligned}\iint_D \operatorname{div} F dA &= \iint_D \left(\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} \right) dA = \int_C (-F_y i + F_x j) \cdot \vec{ds} \\ &= \int_t (-F_y(t)x'(t) + F_x(t)y'(t)) dt = \int_t (F_x(t)i + F_y(t)j) \cdot (y'(t) - x'(t)) dt \\ &= \int_t \vec{F}(t) \cdot \vec{n}(t)r'(t) dt = \int_C \vec{F} \cdot \vec{n} ds\end{aligned}$$

Let $f(x, y) = x^2 + y^3$. We compute $\frac{\partial f}{\partial x} = 2x$ and $\frac{\partial f}{\partial y} = 3y^2$. At $(x, y) = (0, 0)$, we have $\nabla f = (0, 0)$, so $(0, 0)$ is a critical point. Let $(a, b) \times (c, d)$ be an open neighborhood of $(0, 0)$. Then f is minimal when $y \rightarrow c$ and $x = 0$, so that $f \rightarrow c^3 < 0 = f(0, 0)$. Hence, $(0, 0)$ is not a local minimum.

f is maximal when $y \rightarrow d$ and $x \rightarrow \max(-a, b)$, so that $f \rightarrow \max(a^2, b^2) + d^3$. Hence $(0, 0)$ is not a local maximum.

In the x -direction, we have $f(x, 0) = x^2$, which is a minimum. In the y -direction, we have $f(0, y) = y^3$, which is not a max.

Let $f(x, y) = x^2 - y^2$. We compute $\nabla f = \langle 2x, -2y \rangle$, which vanishes at $(0, 0)$. Let $(a, b) \times (c, d)$ be an open neighborhood of $(0, 0)$. f is minimal when $x = 0$ and $y \rightarrow \max(-c, d)$. f is maximal when $x \rightarrow \max(-a, b)$ and $y \rightarrow 0$. Hence, $(0, 0)$ is a saddle point.

Derivative Test. $D_{xx}f = 2, D_{yy}f = -2, D_{xy}f = D_{yx}f = 0$ at $(0, 0)$. Then $\det M = -4 < 0$. Thus, $(0, 0)$ is a saddle point of f .

Consider $f = x^4 + y^4$. Clearly f has a local (global) minimum at $(0, 0)$. However, $D_{xx}f = D_{yy}f = D_{xy}f = 0$, so $\det M = 0$. f can also be taken to be $-x^4 - y^4$, so no results.

Consider $f = x^4 - y^4$. f has a saddle point at $(0, 0)$. However, $D_{xx}f = D_{yy}f = D_{xy}f = 0$, so the derivative test is inconclusive. Indeed, let $(a, b) \times (c, d)$ be a neighborhood of $(0, 0)$. f is maximal with value $\max(a^4, b^4)$ and minimal with value $-\max(c^4, d^4)$, neither of which are zero. Hence, $(0, 0)$ is neither a local min nor max (these conditions are tested given ∇f vanishes).

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_0^1 \frac{x^n}{1+x^n} dx &\leq \lim_{n \rightarrow \infty} \int_0^1 \frac{x^{n-1}}{1+x^n} dx = \lim_{n \rightarrow \infty} \frac{1}{n} \log(1+x^n) \Big|_0^1 \\ &\leq \lim_{n \rightarrow \infty} \frac{\log 2}{n} = 0. \end{aligned}$$

Since $\frac{x^n}{1+x^n} \geq 0$ for $x \in I$, it follows that $\lim_{n \rightarrow \infty} \int_0^1 \frac{x^n}{1+x^n} dx = 0$. f_n converges pointwise to the function

$$f(x) = \begin{cases} 0 & \text{if } x \in [0, 1), \\ \frac{1}{2} & \text{if } x = 1. \end{cases}$$

Let R be a with identity such that the only left ideals are $0, R$. Then R is a division ring.

Proof: Let $r \in R$ be nonzero. Since $[r] \neq 0$, we have $[r] = R$. It follows there exists $s \in R$ such that $sr = 1$, so left inverses exist, ie R is a division ring.

Claim: Let $K \neq \emptyset$ be a subset of $\mathbb{R}^n$. Suppose every $f : K \rightarrow \mathbb{R}$ is bounded. Then K is compact.

Proof: Let (x_n) be a bounded sequence in K . Let $\pi_i : \mathbb{R}^n \rightarrow \mathbb{R}$ denote the projection onto the i th coordinate. $\pi_1(\{x_n\})$ is bounded.

Let X be a set of points with metrics $d_1, d_2 : X \times X \rightarrow \mathbb{R}_{\geq 0}$ such that $d_1(x, y) \leq cd_2(x, y)$ for some $c > 0$. Suppose $x_n \rightarrow x_\infty$ in d_2 for some $x_\infty \in X$. Then $x_n \rightarrow x_\infty$ in d_1 as well.

Proof: Let $\epsilon > 0$. Pick $N \in \mathbb{Z}^+$ such that $n \geq N \rightarrow d_2(x_n, x_\infty) < \epsilon/c$. Then for $n \geq N$, we have

$$d_1(x_n, x_\infty) \leq cd_2(x_n, x_\infty) < c(\epsilon/c) = \epsilon,$$

so $x_n \rightarrow x_\infty$ in d_1 as well.

Let (X, d) be a metric space and $E \subseteq X$. Let $x_0 \in X$. Then the following are logically equivalent.

- (a). x_0 is an adherent point of E .
- (b). x_0 is an interior point of E or a boundary point of E .
- (c). There exists a sequence (x_n) in E which converges to x_0 with respect to the metric d .

(a), (b) are equivalent. Let x_0 be an adherent point of E , so for all $r > 0$, we have $B_r(x_0) \cap E \neq \emptyset$. Suppose FTSOC that x_0 is an exterior point of E . Then there exists $r > 0$ such that $B_r(x_0) \cap E = \emptyset$, absurd. Conversely, if

$x_0 \in E^\circ$, then $x_0 \in E$ and is therefore an adherent point. If x_0 is a boundary point, then in particular for all $r > 0$ we have $B_r(x_0) \cap E \neq \emptyset$, ie x_0 is an adherent point.

(a), (c) are equivalent. For all $n \in \mathbb{Z}^+$, pick $x_n \in B_{1/n}(x_0) \cap E$ (by choice). The sequence (x_n) is in E and converges to x_0 . Conversely, suppose a sequence (x_n) exists. Let $r > 0$. There exists $n \in \mathbb{Z}^+$ such that $1/n < r$. Then $x_n \in B_{1/n}(x_0) \cap E \subseteq B_r(x_0) \cap E$, so x_0 is an adherent point of E .